

Intermolecular Bonds

Intramolecular Bonds are the forces present between atoms, within a molecule.

Intermolecular Bonds are the forces present between molecules.

There are 3 types of intermolecular bonds:

van der Waals Forces:

- dipole-dipole force (DDF)
- London dispersion force (LDF)
- hydrogen bonds

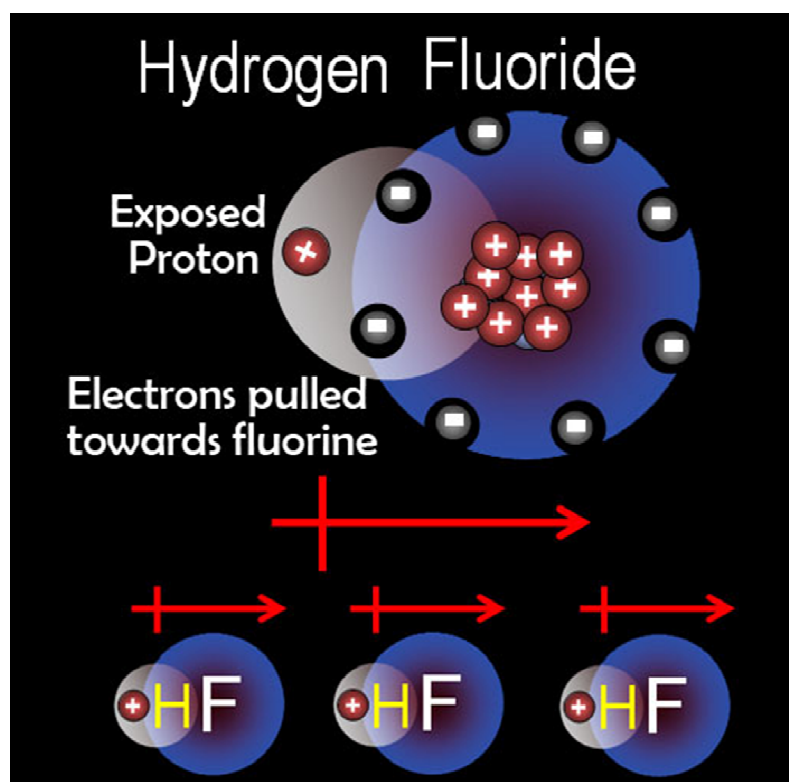
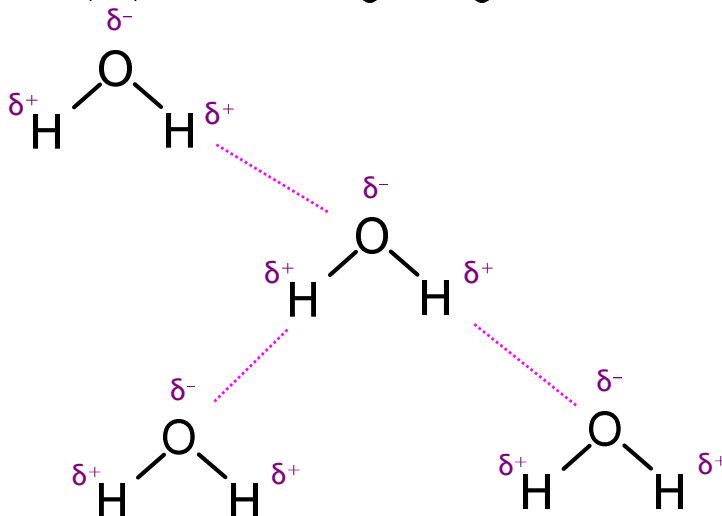
Dipole-Dipole Force (DDF)

This is the force of attraction between polar molecules. The negative end of one molecule is attracted to the positive end of another.



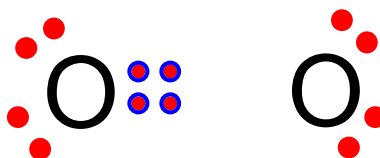
Hydrogen Bonding

Hydrogen bonding occurs between polar molecules that contain H-F, H-O, or H-N bonds. The hydrogen of one molecule forms a direct bond with the F, O, or N of the neighboring molecule. We see this in water:



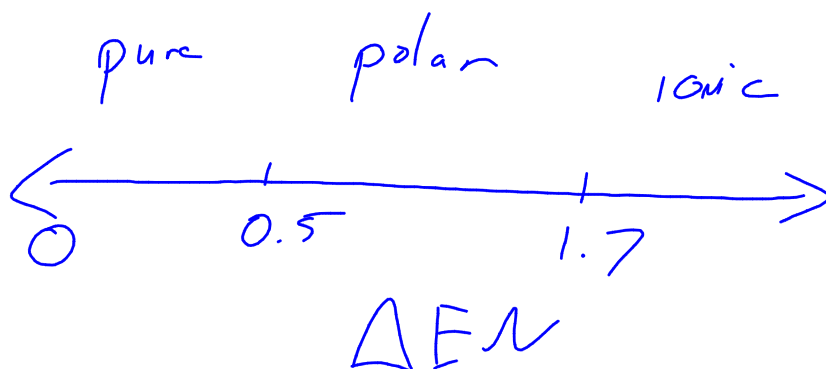
London Dispersion Force (LDF)

This is the force of attraction between non-polar molecules as a result of a temporary imbalance in charge. These temporary imbalances make the molecules appear momentarily polar.



Of the three intermolecular forces, H-bonding is the strongest, while LDF is much weaker. To put it in perspective, the dissociation energy of various bonds is shown below:

Bond Type	Dissociation Energy (kcal)
Covalent bonds	400
Hydrogen Bonds	12-16
DDF	0.5-2
LDF	<1



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Do pg 63 #10-12, 14

Unit Review:

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